

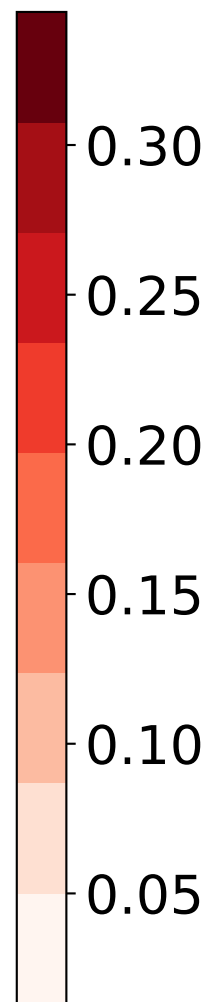
# S10 Step 1 — Per-head rollout (Head 4)

$\mu = 0.06667$

$\sigma = 0.083348$

Target token  $i$

|              |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| [CLS]        | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| funny        | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| ,            | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| heart        | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| ##felt       | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.12 | 0.15 |
| and          | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| endless      | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| ##ly         | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.12 | 0.15 |
| entertaining | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.12 | 0.15 |
| from         | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.06 | 0.05 | 0.12 | 0.15 |
| beginning    | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.06 | 0.05 | 0.12 | 0.15 |
| to           | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.06 | 0.05 | 0.12 | 0.15 |
| end          | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.06 | 0.05 | 0.12 | 0.15 |
| .            | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |
| [SEP]        | 0.34 | 0.03 | 0.04 | 0.02 | 0.03 | 0.05 | 0.01 | 0.01 | 0.01 | 0.04 | 0.02 | 0.05 | 0.05 | 0.13 | 0.15 |



[CLS] funny , heart ##felt and endless ##ly entertaining from beginning to end [SEP]

Source token  $j$